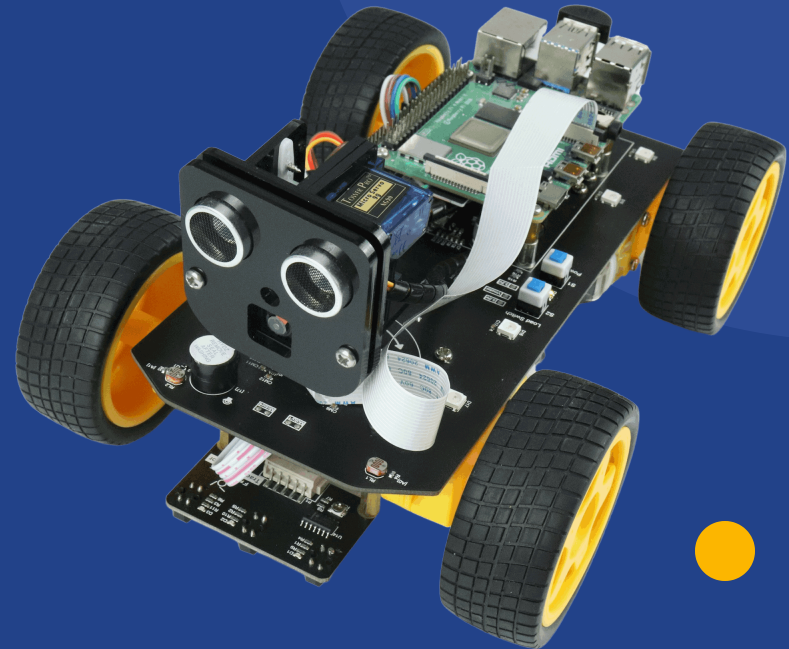




LESSON 9 – FINAL PROJECT WORKSHOP

Machine Learning with Raspberry Pi & Smart Car

Today is **your build day** — combine what you've learned across the course into a smart-car project of your own.





LESSON 9 – FINAL PROJECT WORKSHOP

This is your build day

You work on your **own** final project – and I'm here as your coach.

- Today you work on your **own** final project – start to finish.
- I'm here as your **coach**: ask questions, get unstuck, test ideas.
- **Raise your hand any time** – that's exactly what this session is for.
- Goal by the end: a **working** (or nearly working) project to present next time.



THE DEAL TODAY

The code is **yours**. I'll guide and unblock you – but you drive. Experiment freely.

LESSON 9 – FINAL PROJECT WORKSHOP

Look at everything you can control

Your project can **mix any of these** together – they're all tools in your box now.



LEDs & buzzer – light and sound



Servo head – pan and tilt



Ultrasonic – distance to obstacles



Motors – drive and turn



Commands – client ↔ server



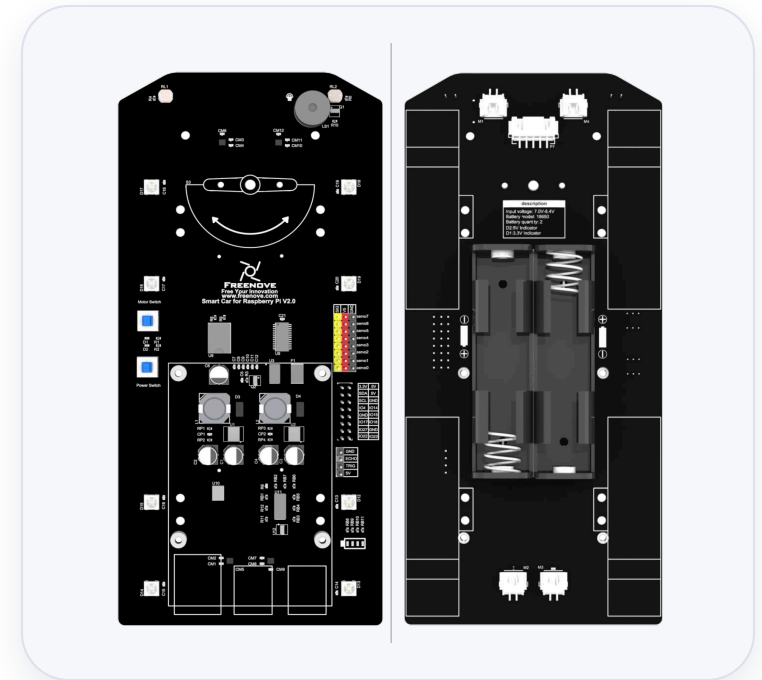
ADC sensors – light & battery



Multithreading – do many things at once



Camera – video & face tracking





LESSON 9 – FINAL PROJECT WORKSHOP

Brainstorm your project

Combine what you've learned into **one idea**. Here are some starters.

Obstacle dodger

Drive, then stop or turn when the **ultrasonic** sees something close.

Light seeker

Use the two **light sensors** to steer toward the brighter side.

Greeter bot

Face tracking + LEDs and buzzer when it spots a face.

Status car

LEDs change color with **battery level** – the watchdog idea.



**Small and working
beats big and broken**

Pick something you can **finish by next class** –
coach is here to help today.



LESSON 9 – FINAL PROJECT WORKSHOP

Scope it right: build in small steps

Combine just **one or two** behaviors you already know – then test as you go.

1 Pick **one** behavior you already know and get it running.

2 **Test it**, then **save** – small steps you can trust.

3 **Only then** add a second behavior on top.

REMEMBER

A finished small project wins

If the car bumps a wall, that's not a bug – it's **data**.
Fix one thing, test, repeat.



LESSON 9 – FINAL PROJECT WORKSHOP

Now go build!

Get your project to a state you can **present next time**.



Save your work as you go



Ask for help whenever you're stuck



Get it ready to **present next time**



NEXT TIME

Final project presentations

Show off what your car can do – you've earned it.
Go build!



• COACH ONLY • HIDDEN SLIDE

How to help during the build

Don't present this slide. Guide and unblock – the project stays theirs.

1

Guide, don't write it for them

Ask **leading questions** and unblock – let the student keep ownership of the code.

2

Push for achievable scope

One or two combined behaviors that actually run – steer away from runaway ideas.

3

Save and test often

Encourage **small steps**: build a little, test, save, repeat. Reframe mistakes as data.



COACH MINDSET

Confidence-building over correctness.
Celebrate every working step.



• COACH ONLY • HIDDEN SLIDE

End of session & after class

Don't present this slide. Wrap up cleanly and set up the presentation day.

1 Every project saved & runnable

Check each student's project is **saved** and will **run** next time before they leave – and make sure they can **finish it and get it running by presentation time**.

2 Remind: next class is presentations

Make sure everyone knows the final project **presentations** are next.

3 Post the final-meeting announcement

Use the [Final Meeting Announcement](#) template on the class page.

EVERY CLASS

✓ Post a class announcement

✓ Introductions if needed

✓ Note absences in #attendance